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Deep Learning

Assignment 2

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1. Introduction

Face verification and one-shot recognition are fundamental problems in deep metric learning and biometric recognition systems. Unlike standard classification tasks, face verification requires determining whether two facial images belong to the same identity, while one-shot recognition requires identifying unseen identities using only a very small number of examples.

Siamese networks and metric-learning approaches are widely used for these tasks because they learn embedding spaces where semantically similar images are mapped closer together and dissimilar images are pushed farther apart. Earlier Siamese approaches commonly relied on binary cross-entropy (BCE) supervision, while more recent methods use metric-learning objectives such as contrastive loss and triplet loss to improve embedding structure and retrieval performance.

In this assignment, we investigate how different metric-learning objectives and backbone architectures affect verification accuracy, embedding quality, and one-shot generalization on the aligned LFW-a dataset. Three experiments were conducted:

- (1) comparison of BCE, contrastive, and triplet-loss objectives,
- (2) comparison between a modified Koch-style CNN and ResNet-18 trained from scratch,
- and (3) evaluation of a frozen pretrained ImageNet ResNet-18 baseline.

Models were evaluated using verification accuracy, ROC-AUC, one-shot classification accuracy, convergence behavior, computational complexity, and qualitative embedding analysis using t-SNE and distance distributions.

2. Dataset

We used the aligned LFW-a (Labeled Faces in the Wild-a) dataset. The dataset contains face images collected under unconstrained real-world conditions, including variations in pose, illumination, facial expression, occlusion, background, and image quality. The aligned version reduces geometric variability while preserving the natural difficulty of the face-verification task.

Training and evaluation followed the provided pairsDevTrain and pairsDevTest protocols. Importantly, identities appearing in the test split were excluded from training, ensuring evaluation on previously unseen individuals rather than memorization of training identities.

The dataset was used:

- in pair-based form for verification experiments
- in episodic N-way form for one-shot evaluation

Positive pairs contain two images of the same identity, while negative pairs contain images belonging to different identities.

To ensure fair comparison across all experiments:

- the same train/validation/test splits were used for every model
- the same one-shot evaluation episodes were reused across configurations
- the same preprocessing and augmentation pipeline was applied throughout all experiments

The distribution of images per identity is highly imbalanced. Some individuals appear many more times than others, while many identities contain only a small number of images. This long-tail distribution increases the difficulty of metric learning because some identities provide limited intra-class variation during training.

The Dataset Statistics and Split Summary is displayed in Table 1.

3. Compute Budget and Fairness Constraints

To ensure fair comparison between configurations, all experiments were conducted under a shared compute budget and consistent evaluation protocol

All configurations in Experiments 1 and 2 used:

- the same train, validation, and test splits, the same aligned LFW-a preprocessing pipeline, the same optimizer (Adam), the same batch size, the same number of training epochs (30) with the same early stopping configuration, the same one-shot evaluation episodes, the same verification evaluation procedure and threshold-selection protocol

Hyperparameter tuning was intentionally limited to a small number of validation-based trials per configuration to maintain comparable search effort across models. Margin selection for contrastive and triplet losses was performed using validation accuracy while keeping all other training conditions fixed.

For Experiment 2, additional fairness constraints were applied to isolate the effect of backbone architecture rather than raw model capacity. The original Koch-style network was modified by reducing the size of its large fully connected layer, resulting in parameter counts within the required 20% range relative to ResNet-18.

All models were trained and evaluated on the same hardware environment.

Each configuration was allowed a maximum of $K=3$ hyperparameter trials for margin selection and learning-rate adjustment.

4. One-shot Evaluation Protocol

One-shot classification performance was evaluated under increasingly difficult N-way settings. All methods exhibited decreasing accuracy as the number of candidate identities increased. A visual comparison of one-shot accuracy across N-way settings is provided in Figure 11.

The BCE-based verification model generalized poorly to one-shot retrieval, with performance dropping from 68.0% in 2-way classification to only 13.0% in 20-way classification. This indicates that BCE optimization learned pairwise decision boundaries rather than a robust embedding space.

Metric learning approaches generalized substantially better. Contrastive loss improved retrieval performance across all settings, while

triplet-based methods achieved the strongest overall results.

Triplet semi-hard mining achieved the best performance in the most difficult 20-way setting (49.5%), suggesting improved discriminative structure in the learned embedding space under challenging retrieval conditions.

Overall, the one-shot evaluation demonstrates that metric-learning objectives produce substantially more transferable embeddings than direct BCE verification training. The one-shot evaluation results are shown in Table 3.

5. Experiment 1

We compare three loss functions while keeping the backbone fixed using a Koch-style Siamese CNN with 128-dimensional L2-normalized embeddings:

1. Koch-style BCE baseline:
shared CNN → elementwise L1 distance → sigmoid → binary cross-entropy
2. Contrastive loss:
Euclidean (L2) distance in embedding space with margin-based supervision

3. Triplet loss:

anchor-positive-negative optimization using semi-hard negative mining, with random triplet sampling reported as an ablation baseline

All configurations used: the same train/validation/test splits, the same backbone architecture, the same optimizer and batch size, the same maximum training budget (30 epochs), identical validation-based early stopping (patience = 5, min_delta = $1e-4$) and the same one-shot evaluation episodes (N=2, N=5, N=20)

Validation was created using a fixed-seed 80/20 pair-level split of pairsDevTrain. Validation was used only for threshold selection, margin selection, and checkpoint selection. Final evaluation used pairsDevTest, whose identities are disjoint from training identities.

Verification Accuracy (validation-selected threshold) is shown in Table 2.

The Roc -AUC and one shot evaluation are also presented in Table 3 and 4 respectively.

Effect of loss function

BCE supervision produced weaker embedding transferability compared with metric-learning objectives.

Contrastive loss substantially improves both ROC-AUC and one-shot accuracy. By directly optimizing Euclidean distances in embedding space, it produces tighter intra-class clustering and improved inter-class separation compared to BCE supervision.

Triplet loss further improves difficult one-shot retrieval tasks. Unlike contrastive loss, triplet loss enforces a relative ordering constraint:

$$d(a, p) + m < d(a, n)$$

which encourages anchor-positive pairs to remain closer than anchor-negative pairs by at least a margin m . This ranking-based objective appears especially beneficial for harder one-shot matching scenarios.

Margin tuning analysis

To investigate the effect of the margin hyperparameter, multiple margins were evaluated for both contrastive and triplet losses. The margin-ablation results are summarized in Table 4 (Appendix). For contrastive loss, performance improved when increasing the margin from 0.5 to 1.0, achieving the highest test accuracy of 75.8%. However, a larger margin of 1.5 reduced performance, suggesting that excessive separation constraints negatively affected generalization.

Margins were tuned using validation accuracy.

For contrastive loss for contrastive loss, margins $m \in \{0.5, 1.0, 1.5\}$ were evaluated, with $m = 1.0$ achieving the strongest validation performance.

For semi-hard triplet loss, margins $m \in \{0.1, 0.2, 0.5\}$ were evaluated.

Observed trend

- small margins produced insufficient inter-class separation
- large margins increased optimization difficulty and reduced test generalization

- aggressive margins improved validation separation but sometimes harmed performance on unseen identities

Interestingly, the highest validation accuracy did not always correspond to the best test accuracy. Semi-hard triplet loss with margin 0.2 achieved the highest validation accuracy (0.966), but margin 0.1 generalized better to the test set. This suggests that highly constrained embedding geometry can over-specialize training identities.

Triplet mining strongly affects training behavior.

Random triplet sampling achieved competitive performance but frequently generated easy negatives that contributed to weak gradients once coarse separation had already been learned.

Semi-hard negative mining instead selected negatives satisfying:

$$d(a, p) < d(a, n) < d(a, p) + m$$

These negatives remain informative because they are difficult enough to violate the margin constraint while still being solvable.

Semi-hard mining substantially increased validation embedding separability and improved performance on harder one-shot tasks ($N=20$). During later training epochs, the number of skipped batches increased because fewer valid semi-hard negatives remained, indicating that the embedding space became progressively more structured over training.

Overall, experiment 1 demonstrates that modern metric-learning losses substantially outperform the original Koch-style BCE verification framework.

The BCE baseline achieved the weakest verification and one-shot performance, supporting the limitation discussed in the original Siamese network paper: pairwise binary supervision alone does not necessarily produce globally structured embeddings.

Triplet semihard mining achieved the strongest overall ROC-AUC and one-shot retrieval performance.

Semi-hard mining improved embedding discrimination substantially during training, although larger margins sometimes reduced transferability to unseen identities.

Triplet semihard mining required substantially higher convergence time due to online hard-negative selection, whereas contrastive loss converged fastest as shown in Figure 4.

The corresponding training-loss curves are shown in Figure 2, while validation-loss behavior is shown in Figure 3.

Verification thresholds were selected exclusively using the validation split. For each epoch, cosine-distance thresholds were swept over the validation pairs, and the threshold achieving maximum validation verification accuracy was retained. Final test accuracy was then computed using this fixed validation-selected threshold to avoid test-set leakage.

6. Experiment 2:

Contrary to the naive expectation that a residual architecture would universally outperform a shallow CNN, the results revealed a more complex trade-off between fitting capacity and generalization.

ResNet-18 achieved the highest validation verification accuracy (0.9364), indicating that the deeper residual architecture possessed stronger representational capacity under metric-learning supervision. However, this improvement did not transfer to unseen identities at test time. Despite lower validation accuracy, the modified Koch CNN achieved substantially stronger test verification accuracy (0.777 vs 0.710), suggesting superior generalization.

This discrepancy indicates that the residual architecture may have overfit the validation identities under the limited-data training regime.

The training dynamics further support this interpretation. ResNet-18 continued improving validation accuracy throughout training (Figure 10), while test accuracy fluctuated and remained consistently below the Koch CNN. In contrast, the

shallower Koch backbone converged earlier yet produced more stable test-set behavior.

Several factors contributed to this outcome.

Small-data regime

LFW-a is relatively small for training deep residual architectures entirely from scratch. ResNet-18 typically benefits from substantially larger datasets, longer optimization schedules, and pretrained initialization. Under constrained data conditions, the deeper model likely possessed excess capacity relative to the available supervision signal.

Optimization and overfitting

Residual connectivity improves gradient flow and optimization stability, enabling the network to fit the training and validation distributions effectively. However, stronger fitting capacity does not necessarily imply stronger transferability to unseen identities. The gap between validation and test performance suggests that the deeper residual model learned more identity-specific representations that generalized less effectively.

Task specialization

The original Siamese CNN was explicitly designed for low-resolution pairwise similarity learning. The aligned 105×105 facial images used in this assignment may favor:

- shallow local feature extraction,
- aggressive spatial compression,
- and compact embedding functions

over deeper hierarchical residual representations trained from scratch.

Efficiency-performance trade-off

Despite weaker test generalization, ResNet-18 remained substantially more computationally efficient:

- approximately 2× lower FLOPs,
- substantially faster convergence,
- and lower wall-clock training time as illustrated in Figure 9.

This highlights an important practical trade-off:

Higher-capacity architectures may achieve stronger validation fitting while simultaneously generalizing worse under limited-data metric-learning conditions.

Experiment 2 isolated architectural effects by closely matching parameter counts between a modified Koch CNN and ResNet-18 under identical metric-learning supervision.

ResNet-18 achieved the highest validation accuracy, demonstrating stronger fitting capacity and optimization behavior. However, the modified Koch CNN achieved superior test verification accuracy, indicating stronger generalization to unseen identities.

The results therefore suggest that deeper residual architectures do not automatically produce superior embedding quality when trained entirely from scratch on limited facial datasets.

Under constrained data and training budgets, the simpler Siamese CNN appeared:

- easier to regularize,
- less prone to overfitting,
- and more transferable to unseen identities.

Meanwhile, ResNet-18 achieved substantially lower computational cost and faster convergence, indicating a clear efficiency advantage despite weaker generalization performance.

FLOPs were computed using the convention $\text{FLOPs} = 2 \times \text{MACs}$ using the thop library on 105×105 RGB inputs.

The same verification pairs and one-shot evaluation episodes were reused across both backbone configurations.

These findings suggest that architecture selection in metric learning involves a trade-off between representational capacity, computational efficiency, and generalization robustness.

Training and validation loss curves for both backbone configurations are shown in Figures 7 and 8, respectively.

7. Experiment 3 –

As a transfer-learning baseline, a frozen ImageNet-pretrained ResNet-18 was evaluated without any fine-tuning on LFW-a. The model was used purely as a fixed feature extractor, and cosine similarity was applied for face verification. The same aligned face images, test pairs, and one-shot episodes used

in the previous experiments were reused here to ensure fair comparison across all models.

The pretrained baseline achieved a verification accuracy of 70.1% with an ROC-AUC of 0.774. One-shot accuracies were 77.5% for 2-way classification, 56.0% for 5-way classification, and 32.0% for 20-way classification. Full quantitative results are summarized in Table 8, while the ROC curve is shown in Figure 12.

Despite requiring no task-specific training, the pretrained model performed reasonably well. This suggests that ImageNet-pretrained visual representations already contain useful semantic and structural information that transfers effectively to facial similarity tasks. Even generic pretrained features were able to produce meaningful identity discrimination when combined with cosine similarity.

The frozen pretrained baseline also outperformed the BCE Siamese model from Experiment 1, indicating that large-scale visual pretraining can sometimes provide more transferable representations than weak pairwise BCE supervision alone. Interestingly, its performance was also close to the ResNet18 model trained from scratch in Experiment 2, despite requiring no

optimization on LFW-a. This highlights the importance of pretrained initialization when training data is limited.

However, the pretrained baseline remained clearly weaker than the best metric-learning system obtained in Experiment 1 using semihard triplet loss. In particular, one-shot retrieval performance degraded more noticeably for difficult 20-way tasks, suggesting that generic pretrained embeddings are less discriminative than embeddings optimized specifically for face verification.

No training or validation curves are reported for Experiment 3 since the pretrained ResNet-18 remained fully frozen and no optimization was performed.

Overall, Experiment 3 demonstrates that pretrained visual representations provide a strong baseline for face verification under limited-data conditions, but carefully optimized metric-learning objectives still produce the most effective and transferable facial embeddings.

A combined ROC comparison across all evaluated models is shown in Figure 13.

8. Embedding Analysis

The best overall embedding model was the Koch CNN trained with semi-hard triplet loss. To qualitatively inspect the learned representation, a 2D t-SNE projection was generated using embeddings from 25 sampled identities in the test split, shown in Figure 16.

The t-SNE plot shows partial identity grouping, with several local clusters forming across the embedding space. This indicates that the model learned meaningful facial similarity structure rather than mapping identities randomly. However, the clusters are not fully separated. Several identity groups overlap or appear mixed, especially in dense central regions of the projection. This is consistent with the remaining verification and one-shot errors observed in the quantitative results.

The imperfect separation is expected given the limited dataset size, visual similarity between some identities, and variation in pose, illumination, expression, and image quality. Overall, the t-SNE visualization supports that metric learning improved the embedding structure, but also shows that the learned space is not perfectly class-separable.

Distance Analysis

The embedding-distance distributions provide a clearer quantitative view of the learned metric space, shown in Figure 17. Same-identity pairs had a lower mean Euclidean distance than different-identity pairs:

Distance Type	Mean	Std
Intra-class	0.8222	0.2808
Inter-class	1.3442	0.2646

The intra-class distribution is shifted left relative to the inter-class distribution, confirming that the model generally places images of the same identity closer together than images of different identities.

However, the two distributions still overlap. This overlap explains the remaining verification errors: some different-identity pairs are embedded closer together than some same-identity pairs. These errors are likely caused by visually similar identities, pose changes, illumination variation, facial expression differences, and the limited scale of the training set.

Overall, the distance analysis confirms that the learned embedding space has the intended metric-

learning structure: reduced same-identity distances and increased different-identity distances, although not perfect separation.

Failure case Analysis

Qualitative failure-case examples and corresponding hypotheses are summarized in Table 5, with representative false accepts shown in Figure 18 and false rejects shown in Figure 19.

For the strongest configurations, aggregate training-loss and validation-accuracy trends are summarized in Figures 14 and 15.

9. Threats to Validity

Several limitations should be considered when interpreting these results.

First, all experiments were trained under limited compute budgets and relatively short training schedules. Larger models such as ResNet-18 trained from scratch may not have fully converged, so some comparisons also reflect optimization efficiency under constrained resources.

Second, only a small number of random seeds were evaluated. Additional runs could reveal greater variability in performance and model ranking stability.

Third, the experiments were conducted only on aligned LFW-a face crops. Since LFW-a is relatively small and clean, the results may not generalize to more difficult real-world settings with stronger pose variation, occlusion, illumination changes, or demographic diversity.

Hyperparameter tuning was also intentionally limited for fairness and computational feasibility. Different margins, learning rates, or mining strategies could potentially improve some models further.

Finally, the t-SNE visualizations are qualitative tools and may distort global embedding geometry. They should therefore be interpreted as approximate visual summaries rather than exact representations of the learned feature space.

A complete quantitative comparison across all experiments and baselines is provided in Table 9.

10. Appendix – Additional Figures, Tables and Detailed Results

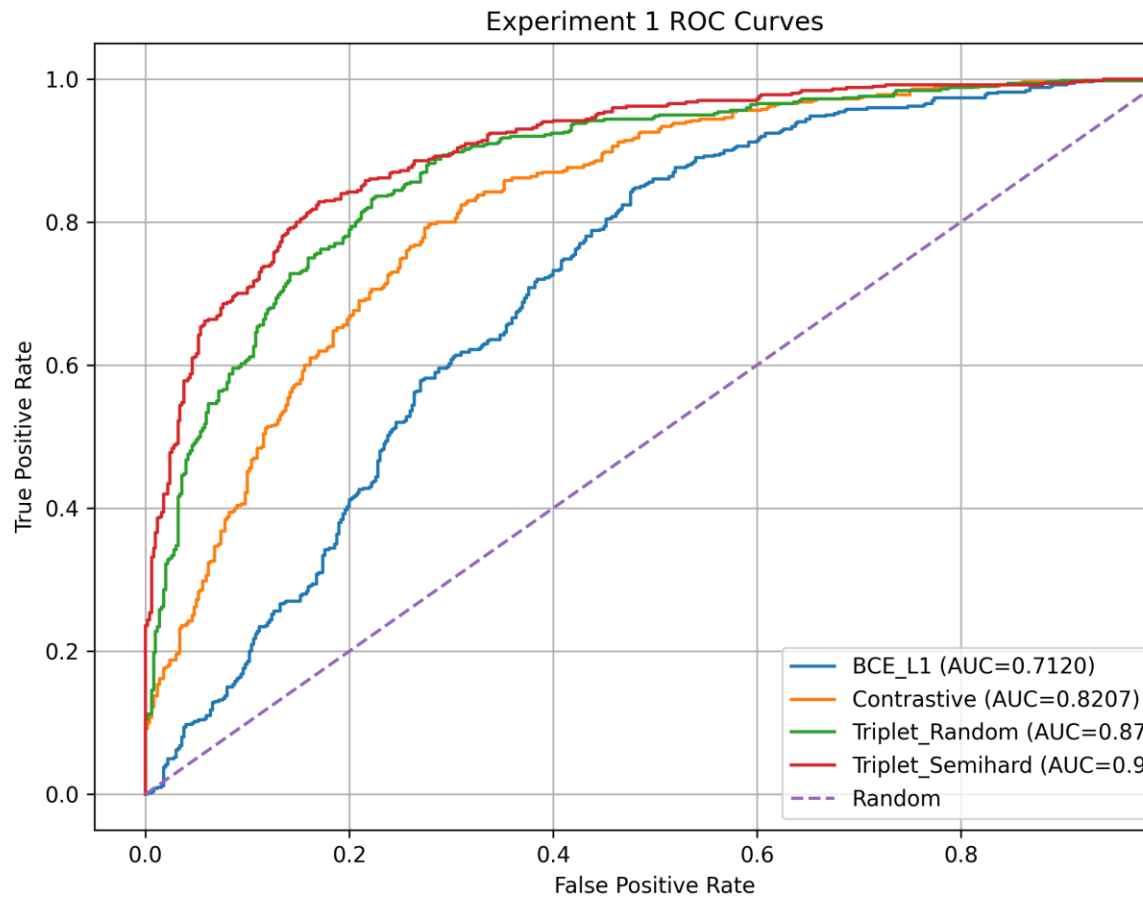
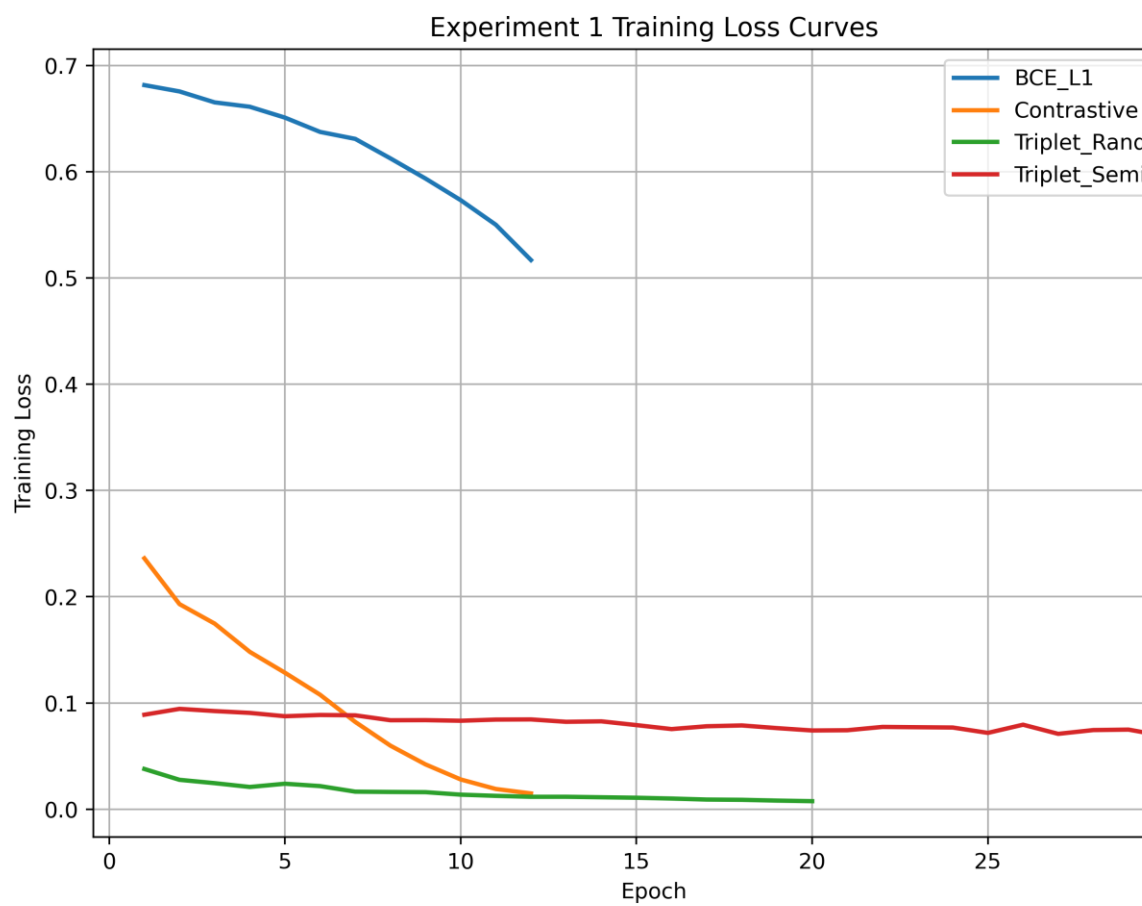


Figure 1: ROC curves for Experiment 1 comparing BCE, contrastive, and triplet objectives under the same backbone and evaluation protocol.

Figure 2: Exp 1 training loss curves



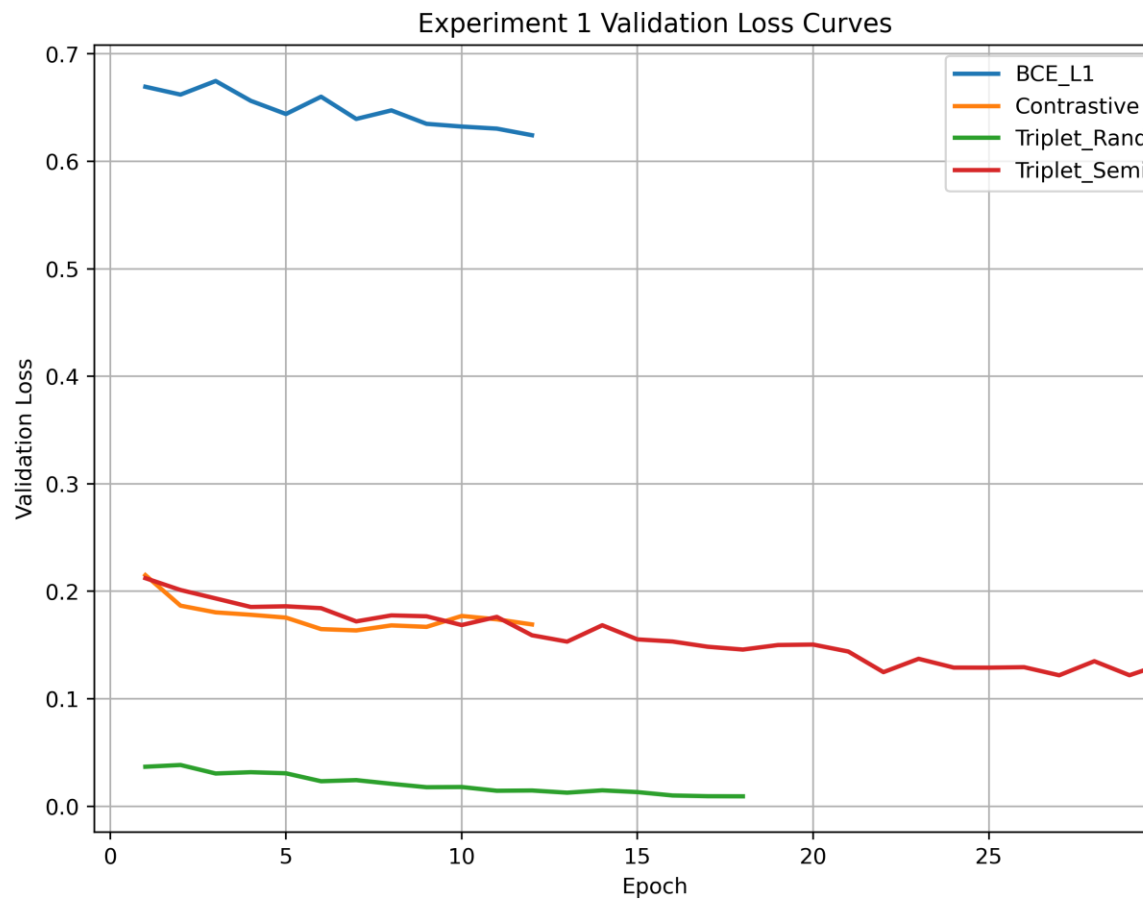


Figure 3: Experiment validation loss curve

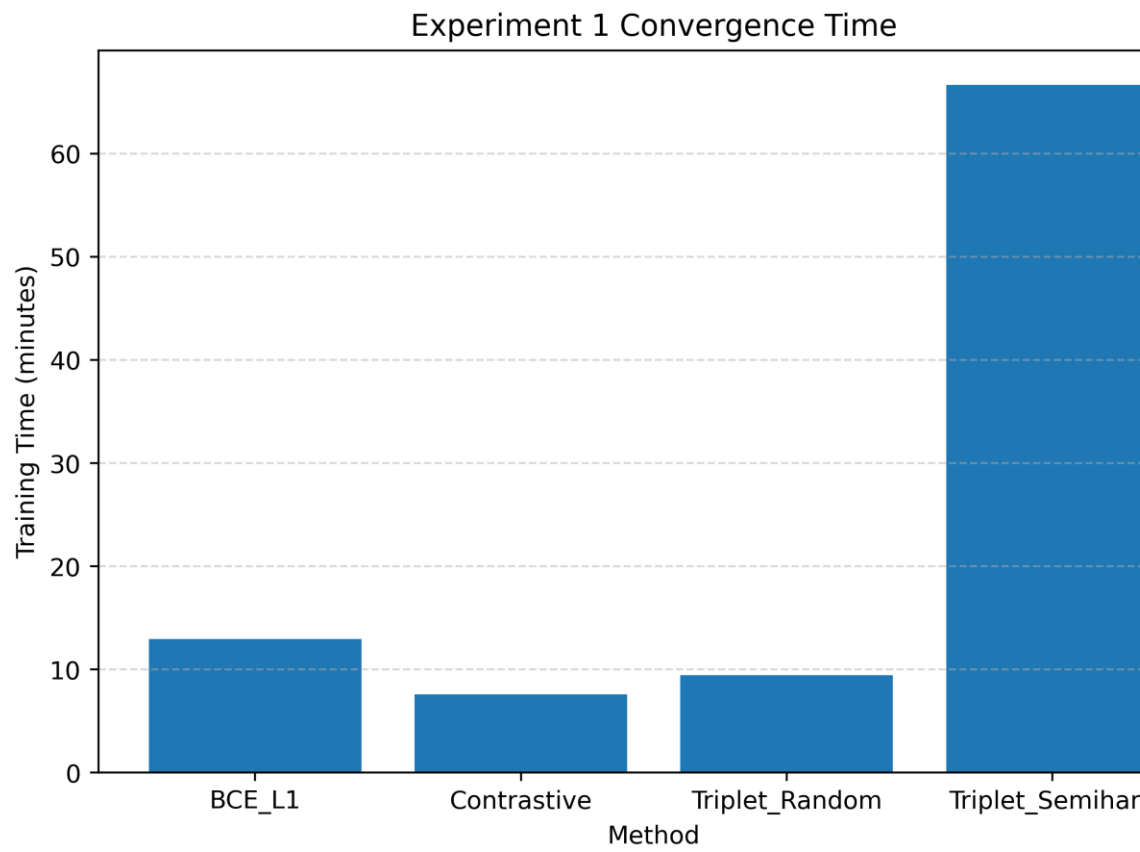


Figure 4: Total convergence training time for Experiment 1 models measured in wall-clock minutes.



Figure 5: Experiment 1 validation accuracy curves

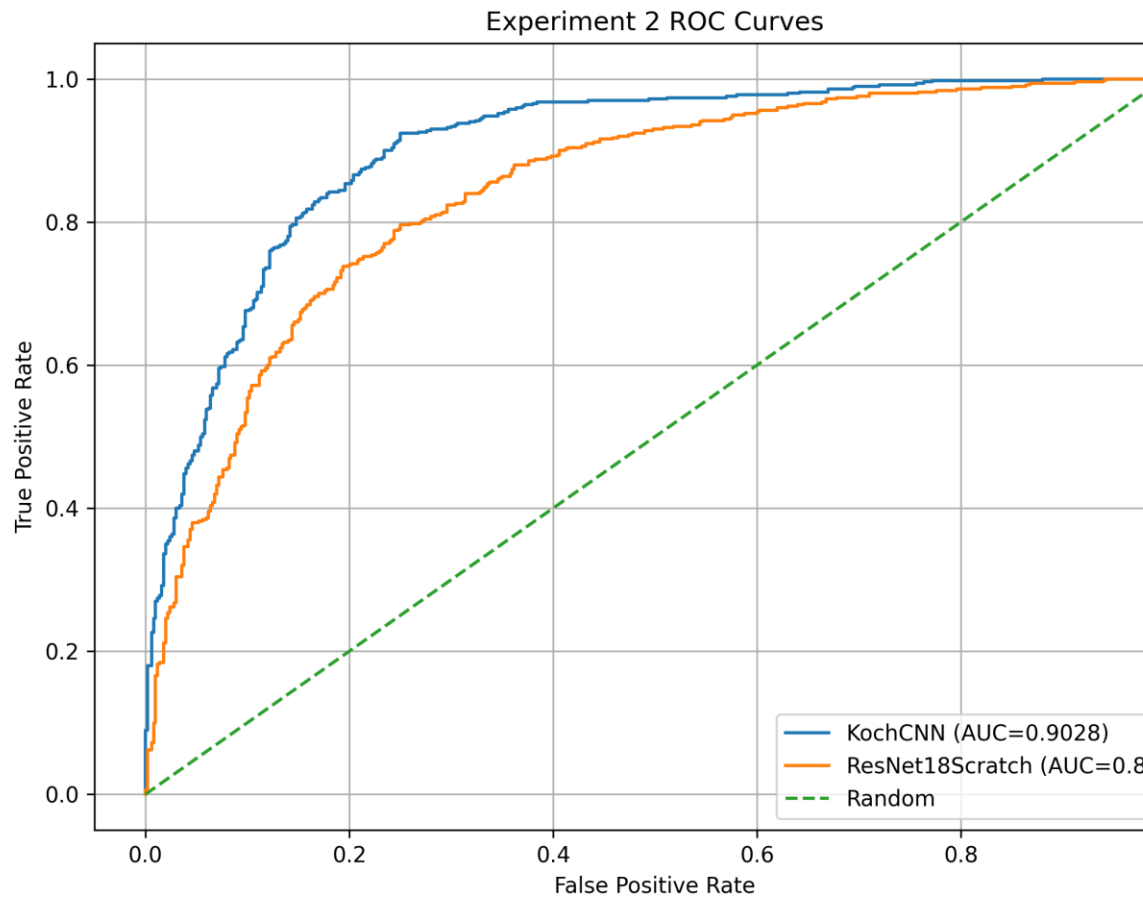


Figure 6: ROC curves for Experiment 2 backbone comparison under matched parameter counts and identical triplet-loss supervision.



Figure 7: Experiment 2 Training loss



Figure 8: Experiment 2 Validation loss

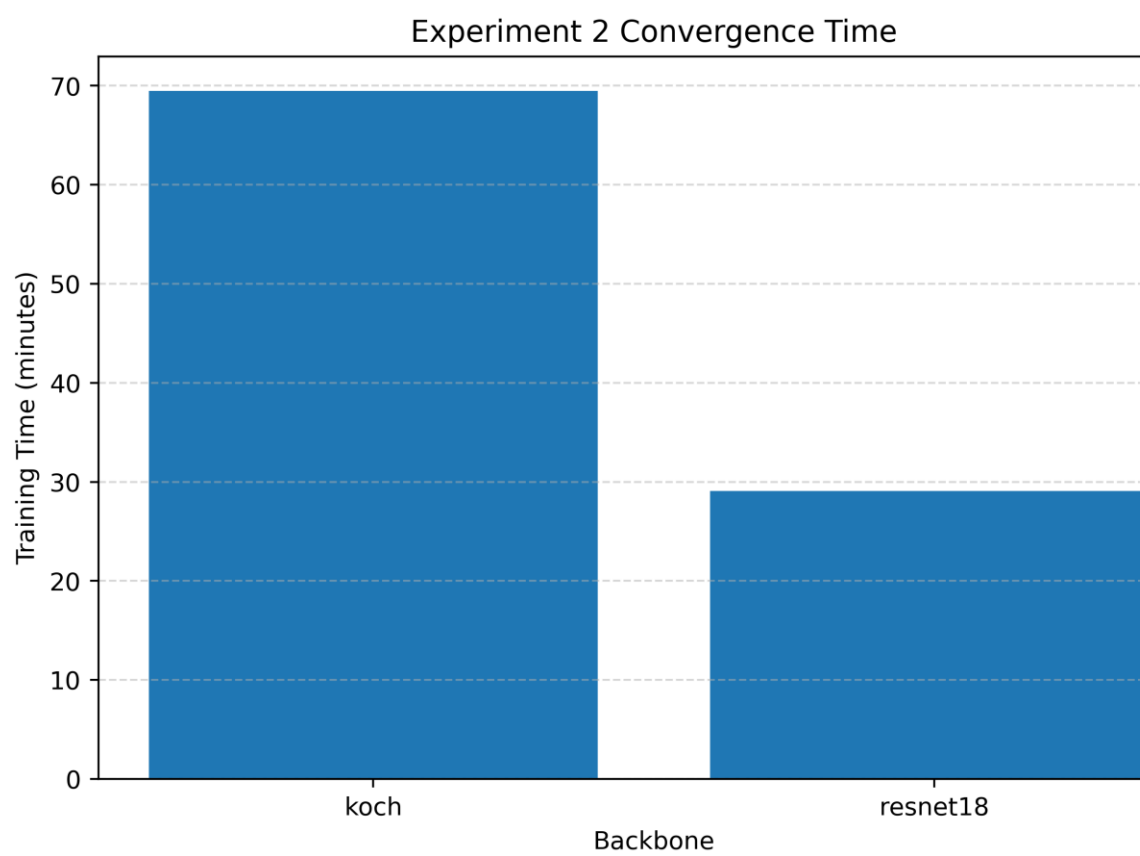


Figure 9: Experiment 2 Convergence time

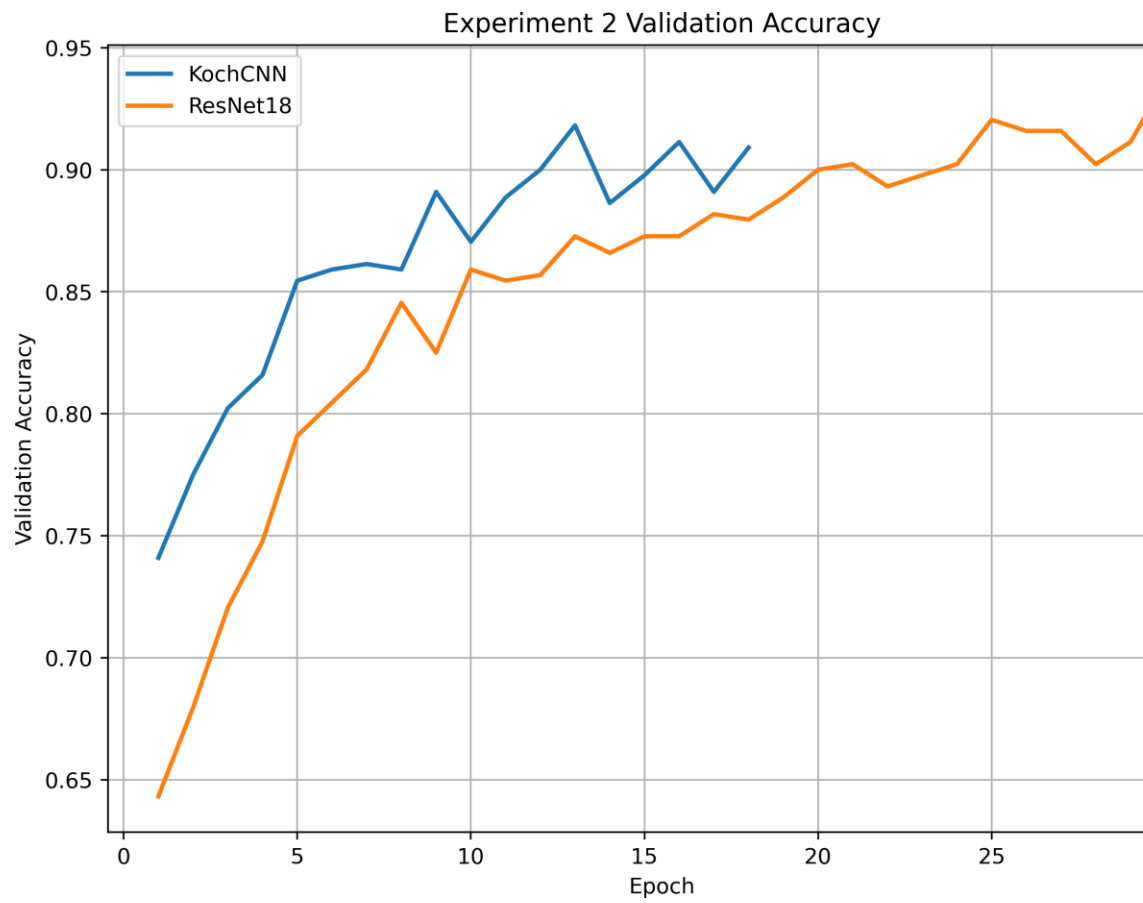


Figure 10: Experiment 2 Validation Accuracy

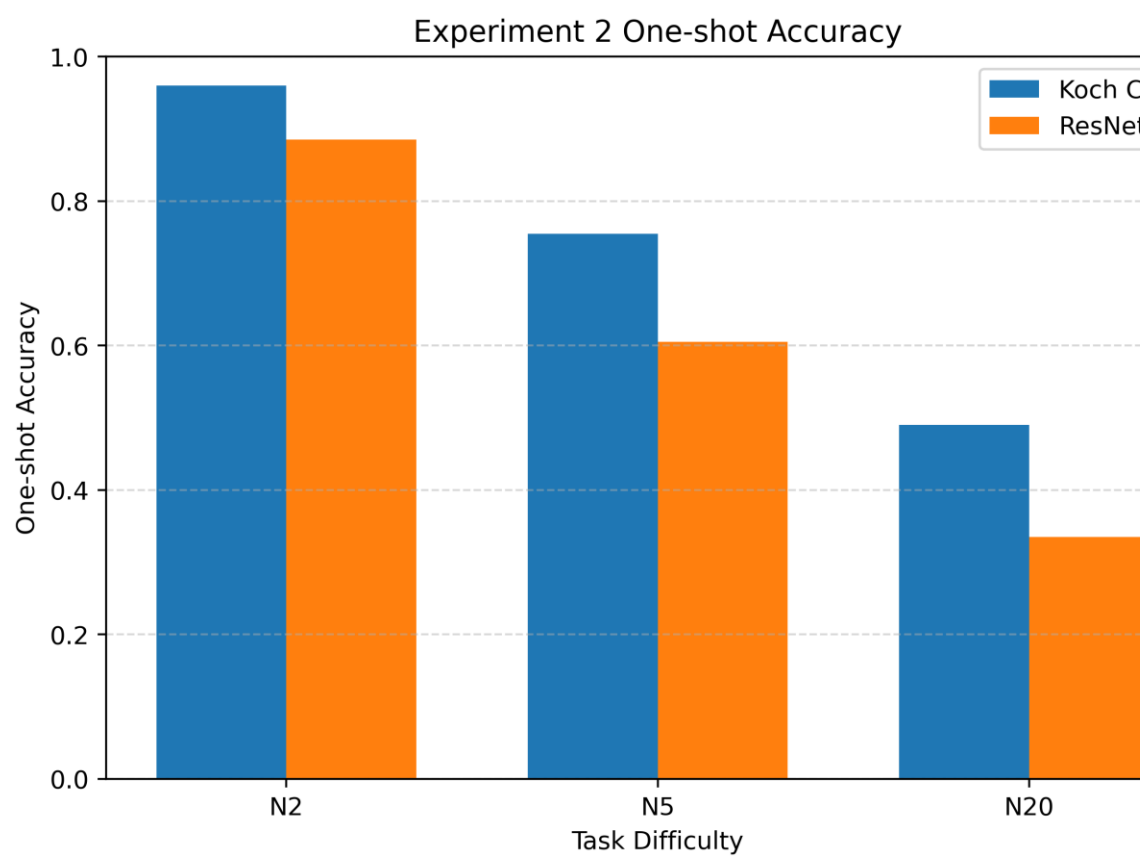


Figure 11: One-shot identification accuracy for Experiment 2 across increasing candidate-set sizes.

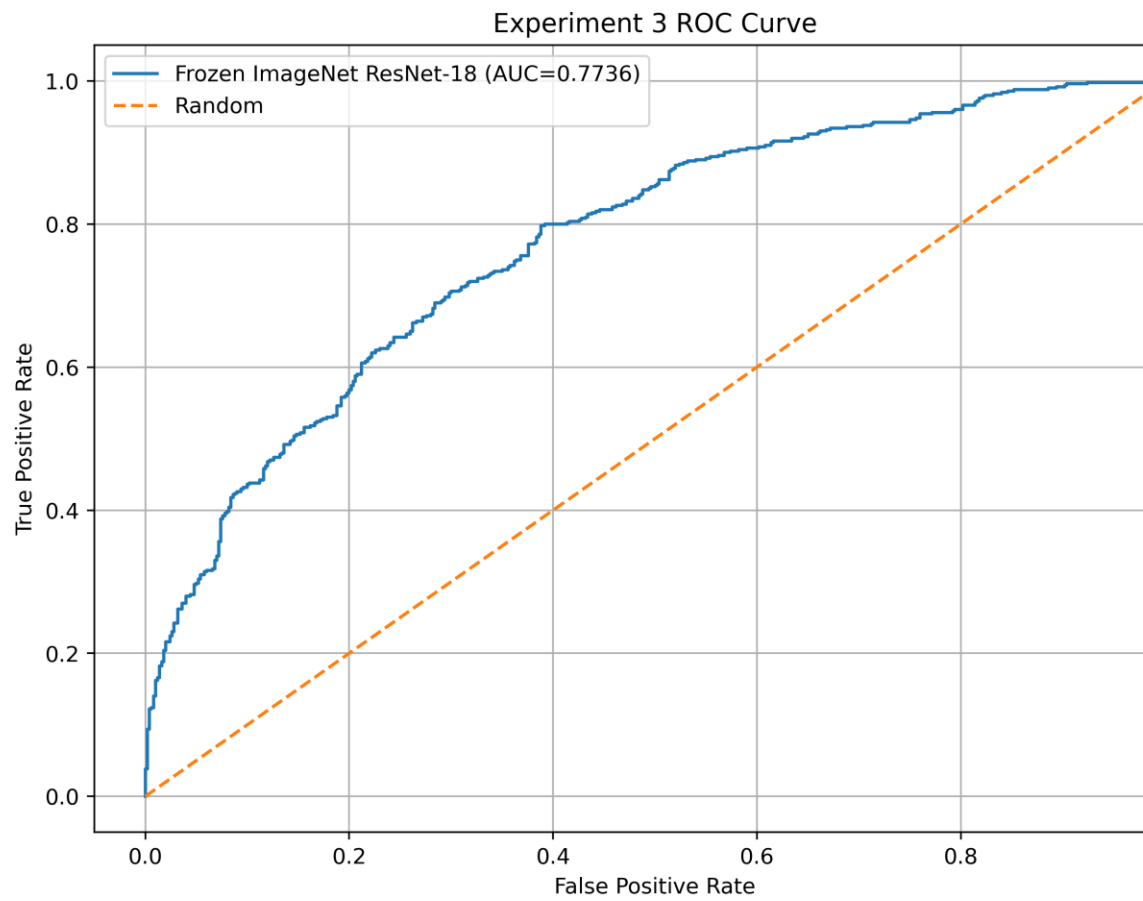


Figure 12: Experiment 3 ROC curve

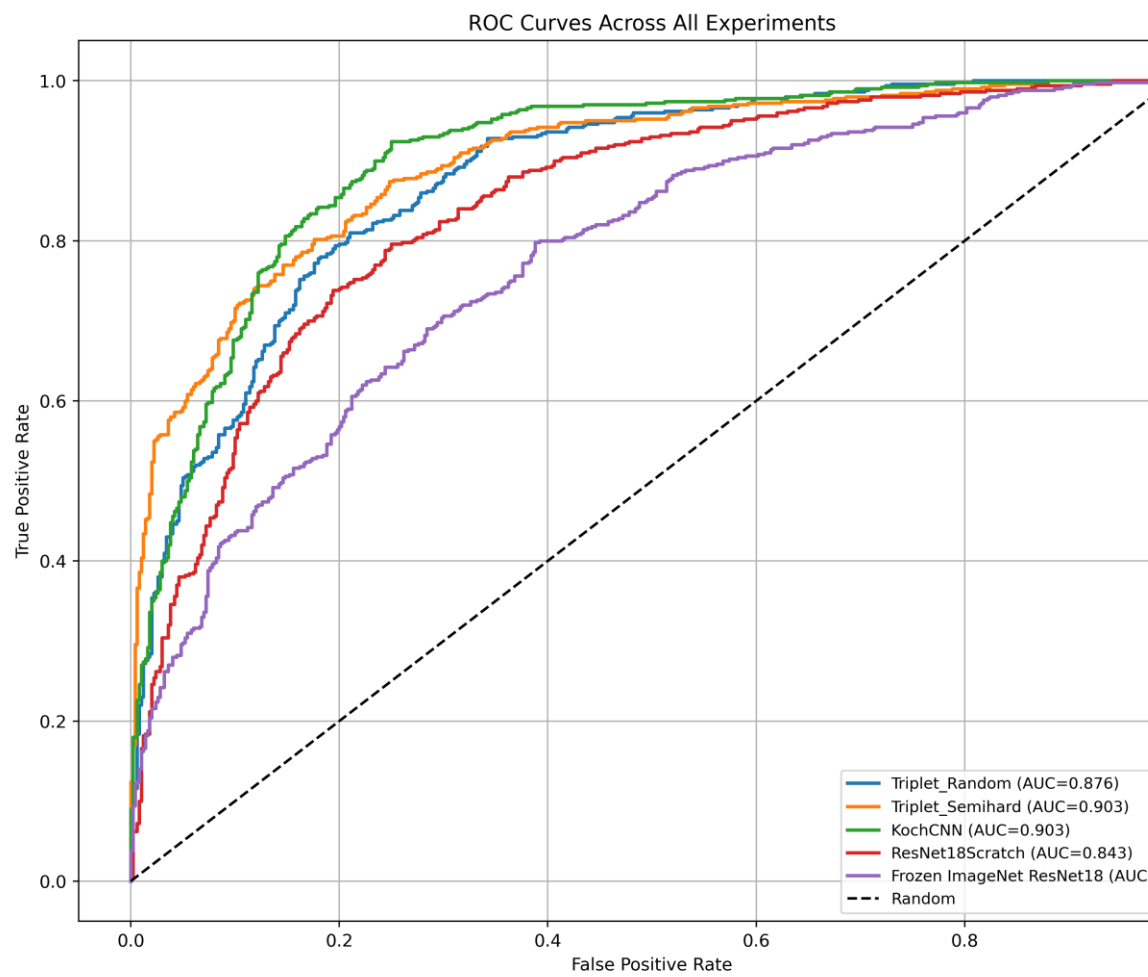


Figure 13: ROC comparison across all models

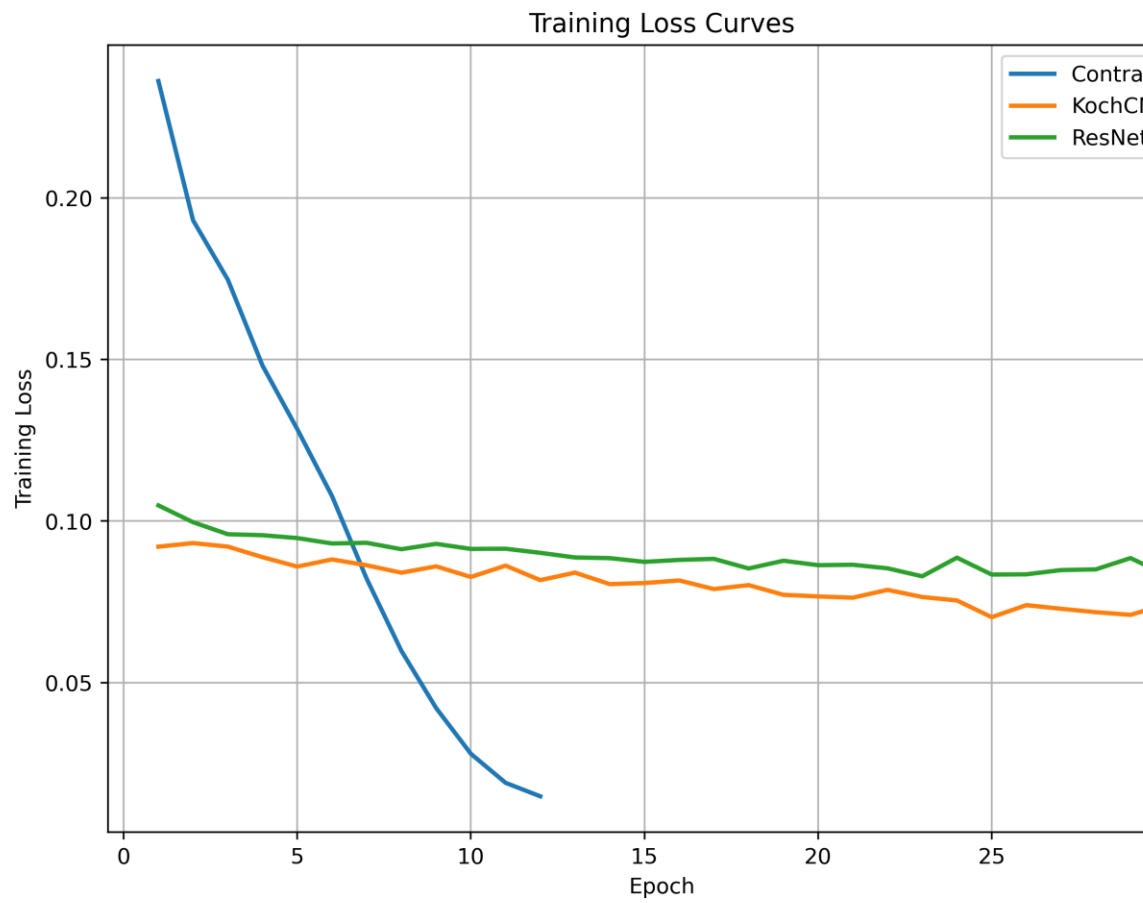


Figure 14: Training loss curve

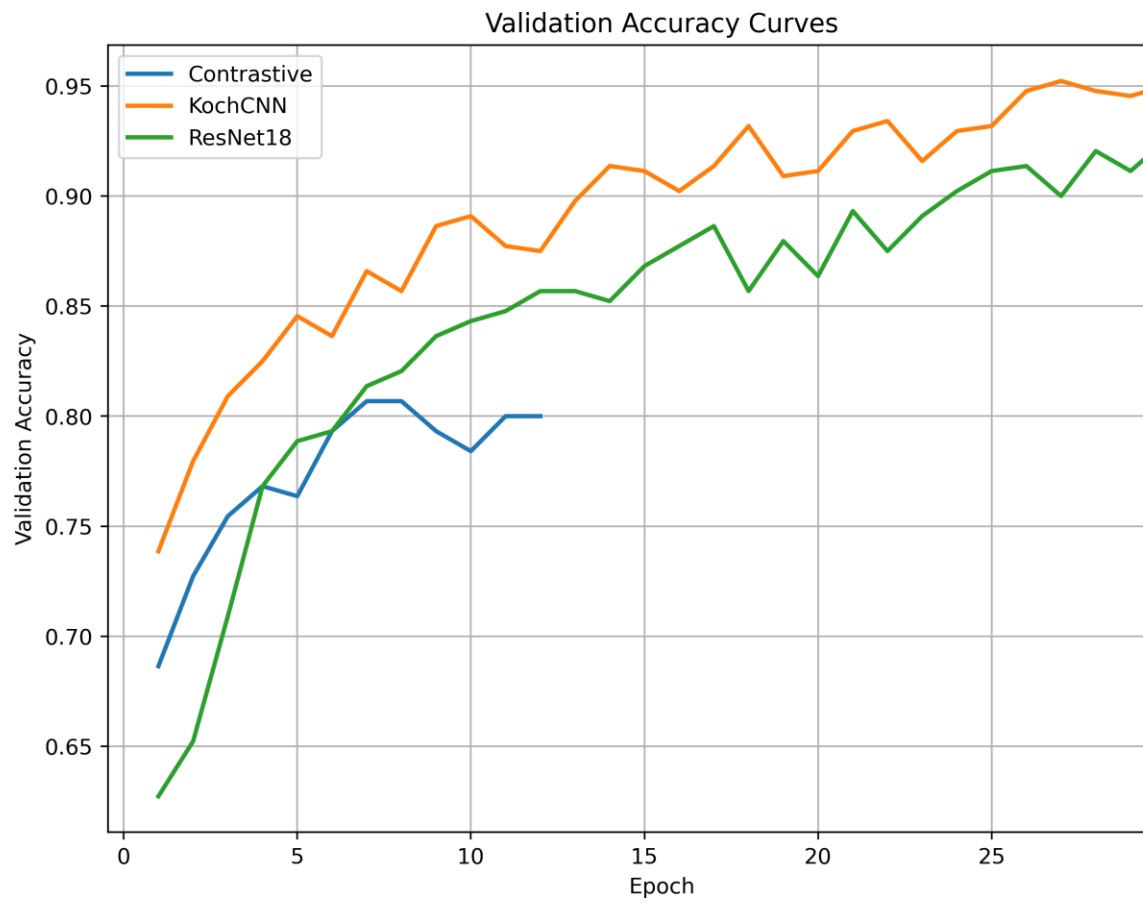


Figure 15: Validation accuracy curves across training epochs for the strongest configurations in Experiments 1 and 2.

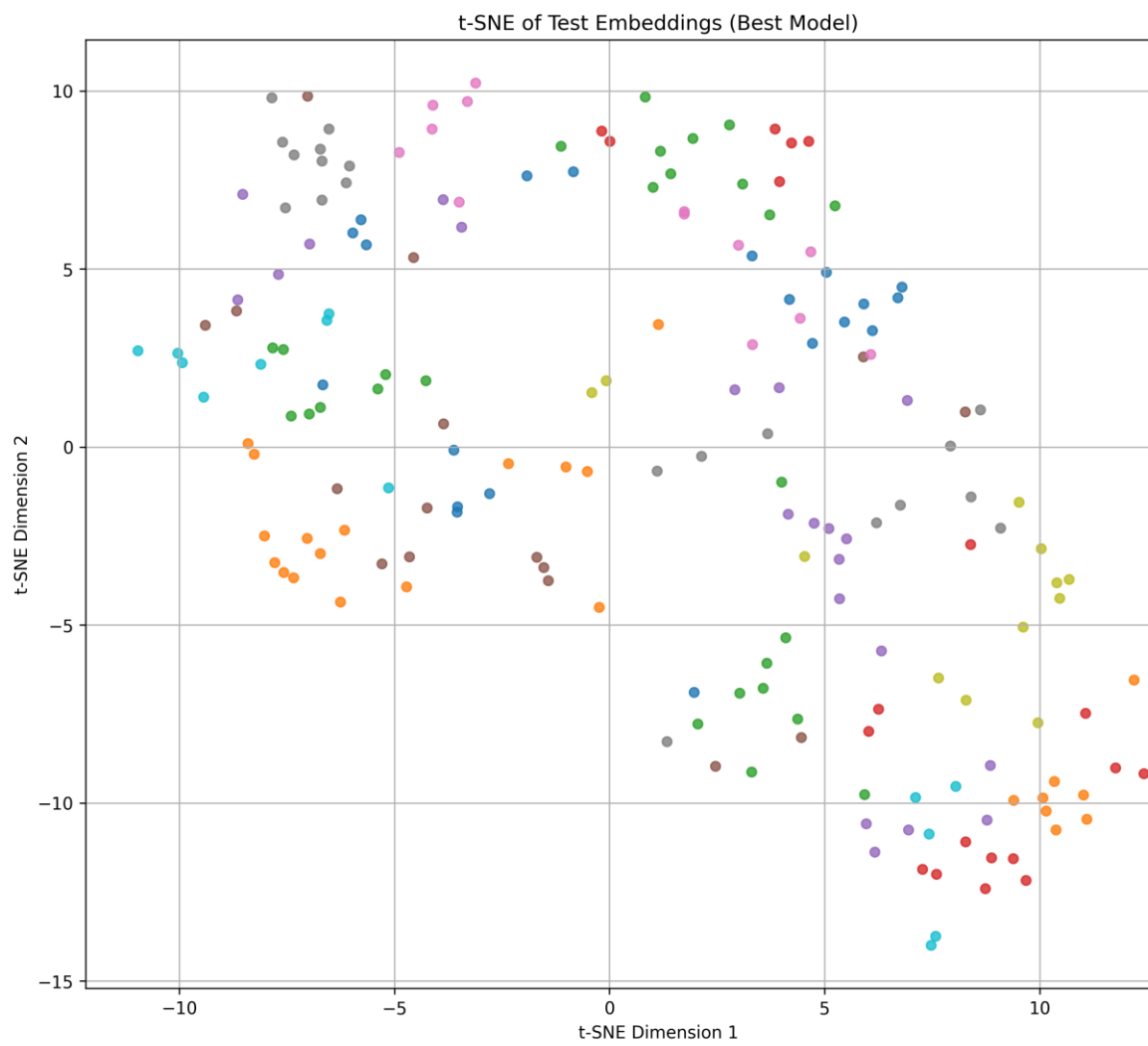


Figure 16: t-SNE projection of test-set embeddings produced by the best-performing model (Koch CNN with semi-hard triplet loss). Twenty-five identities from the test split were sampled for visualization. Several compact identity-specific clusters emerge, indicating meaningful metric structure in the learned embedding space.

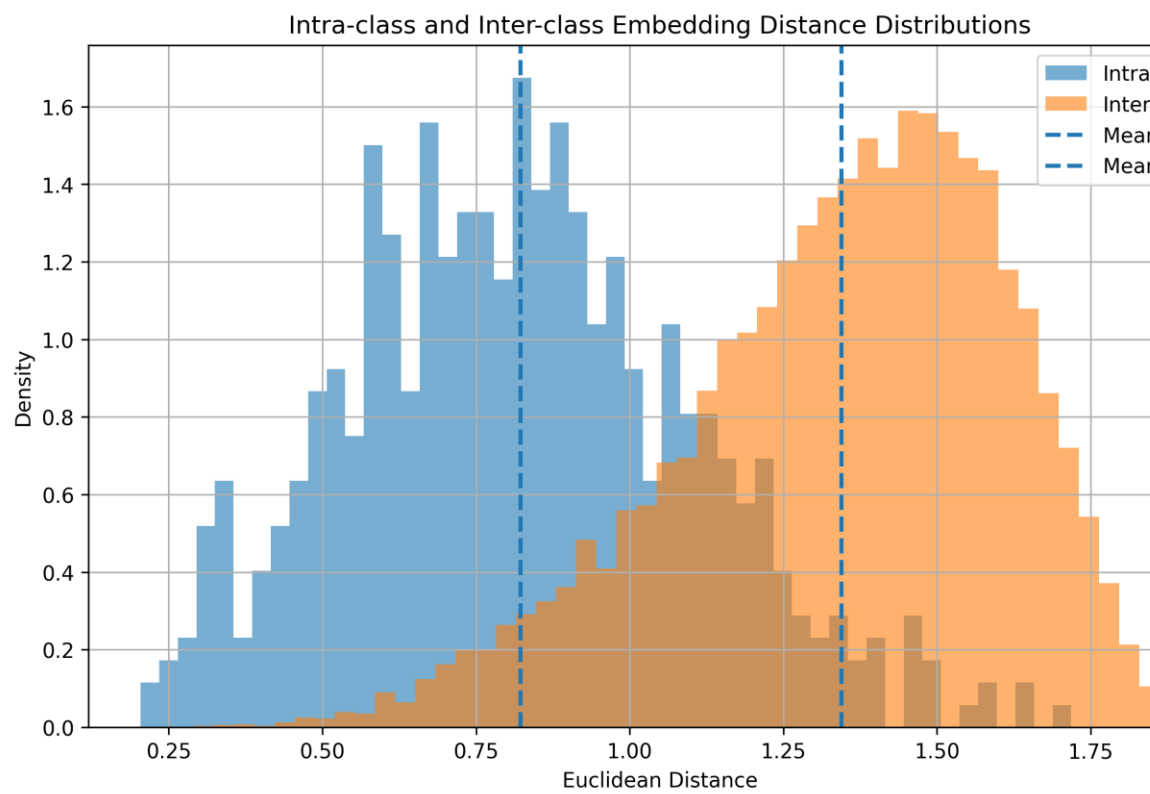


Figure 17: Intra-class and inter-class embedding distance distributions for the best-performing model (Koch CNN with semi-hard triplet loss).

False Accepts

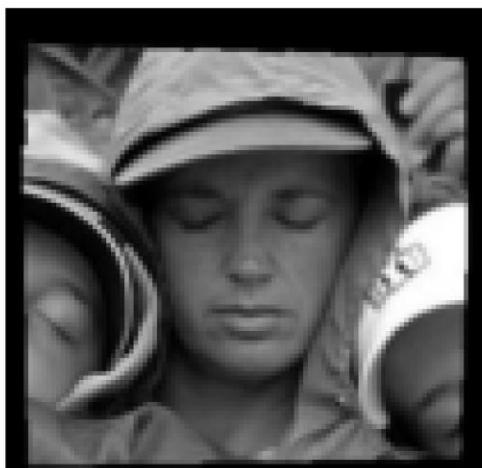
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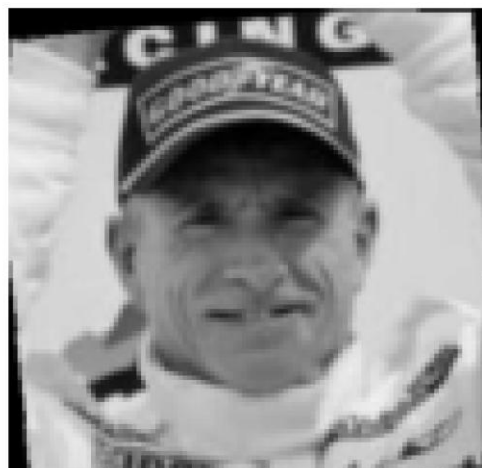
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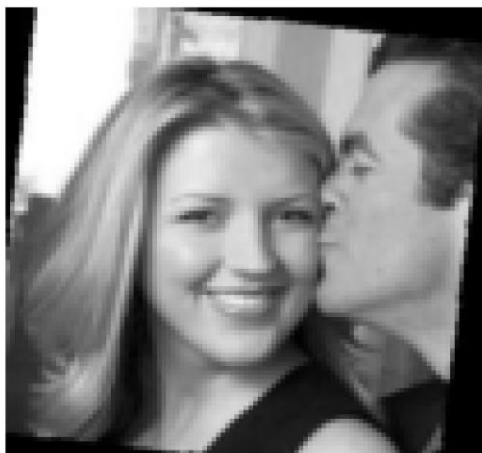
Dist=0.531



Dist=0.531



Dist=0.879



Dist=0.879

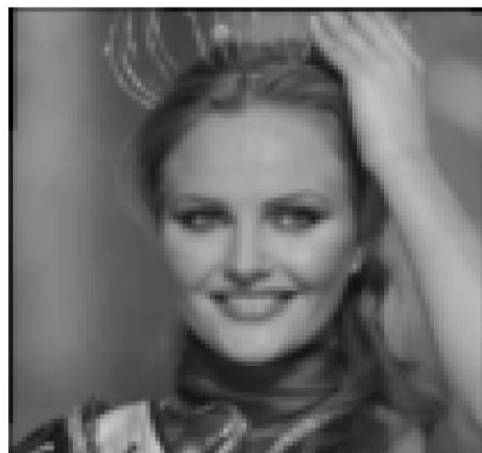


Figure 18: False accepts

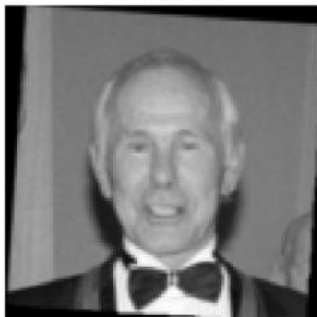
Dist=0.973



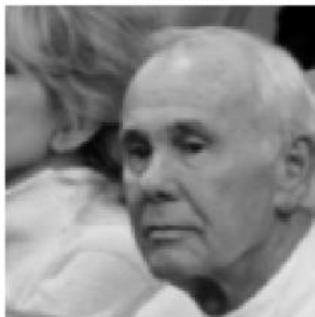
Dist=0.973



Dist=1.488



Dist=1.488



Dist=1.159



Dist=1.159



Dist=1.241

Dist=1.241

Figure 19: False rejections

Table 1: Dataset split

Dataset statistic	Value
Total images	13,233
Total identities	5,749
Min images per identity	1
Max images per identity	530
Mean images per identity	2.30
pairsDevTrain pairs	2,200
pairsDevTest pairs	1,000

Table 2: ROC/ AUC experiment1

Method	AUC
BCE_L1	0.712
Contrastive	0.821
Triplet Random	0.877
Triplet Semi-hard	0.898

Table 3: One-shot Accuracy (same evaluation episodes)

Method	N=2	N=5	N=20
BCE_L1	0.680	0.375	0.130
Contrastive	0.855	0.585	0.325
Triplet Random	0.915	0.770	0.480
Triplet Semi-hard	0.925	0.755	0.495

Table 4: Effect of margin hyperparameter on validation and test accuracy for contrastive and triplet loss models.

Loss Type	Margin	Validation Accuracy	Test Accuracy
Contrastive	0.5	0.802	0.747
Contrastive	1.0	0.807	0.758
Contrastive	1.5	0.791	0.741
Triplet	0.1	0.939	0.806
Triplet	0.2	0.957	0.800
Triplet	0.5	0.952	0.741

Table 5: failure-case analysis

Case	Error Type	Hypothesis
FA-1	False Accept	Similar frontal pose, grayscale intensity, and limited texture detail reduced discriminative cues.
FA-2	False Accept	Occlusion and blur reduced fine-grained identity information, causing reliance on coarse facial structure.
FA-3	False Accept	Similar global appearance and portrait composition caused embeddings to become unusually close.
FR-1	False Reject	Differences in expression, lighting, and sharpness increased embedding distance for the same identity.

Case	Error Type	Hypothesis
FR-2	False Reject	Aging and illumination variation reduced embedding similarity despite identical identity labels.
FR-3	False Reject	Pose and shadow differences increased intra-class distance beyond the verification threshold.

Table 6: Verification and one-shot evaluation results for Experiment 2 backbone comparison.

Backbone	Validation Accuracy	Test Accuracy	AUC	One-shot N=2	One-shot N=5	One-shot N=20
KochCNN	0.918	0.777	0.903	0.960	0.755	0.490

Backbone	Validation Accuracy	Test Accuracy	AUC	One-shot N=2	One-shot N=5	One-shot N=20
ResNet18Scratch	0.936	0.710	0.843	0.885	0.605	0.335

Table 7: Computational complexity and convergence statistics for Experiment 2 models.

Backbone	Parameters	FLOPs	Training Time (min)	Best Epoch
KochCNN	10.78 M	2.00 B	69.46	13
ResNet18Scratch	11.24 M	0.95 B	29.06	30

Table 8: Experiment 3 results

Metric	Value
AUC	0.774
Test Accuracy	0.701
One-shot N=2	0.775
One-shot N=5	0.560
One-shot N=20	0.320

Table 9: Summary Final Table

Experiment	Model	Loss	Margin	Validation Accuracy	Test Accuracy	AUC	N=2	N=5	N=20	Training Time (min)
Exp1	BCE_L1	BCE	—	0.716	0.672	0.712	0.680	0.375	0.130	12.98

Experiment	Model	Loss	Margin	Validation Accuracy	Test Accuracy	AUC	N=2	N=5	N=20	Training Time (min)
Exp1	Contrastive	Contrastive	1.0	0.807	0.758	0.821	0.855	0.585	0.325	7.57
Exp1	Triplet_Random	Triplet	0.2	0.893	0.783	0.876	0.915	0.770	0.480	9.46
Exp1	Triplet_Semihard	Triplet	0.2	0.957	0.800	0.903	0.925	0.755	0.495	66.66
Exp2	KochCNN	Triplet Semihard	0.2	0.918	0.777	0.903	0.960	0.755	0.490	69.46
Exp2	ResNet18Scratch	Triplet Semihard	0.2	0.936	0.710	0.843	0.885	0.605	0.335	29.06
Exp3	Frozen ImageNet ResNet18	None (Frozen Features)	—	—	0.701	0.774	0.775	0.560	0.320	N/A